1. $S \to YC \mid AZ$

$Y \to aYb \mid \lambda$

$Z \to bZc \mid Zc \mid \lambda$

$A \to aA \mid \lambda$

$C \to cC \mid \lambda$

$$\begin{array}{ccccccccc} & & & & 1 & 2 & 3 & 4 & 5 \\ S \to & YC & \mid & AZ & \mid & Y & \mid C & \mid A & \mid Z \end{array}$$

$Y \to aYb \mid ab$ (8)

$Z \to bZc \mid Zc \mid bc \mid c$ (6, 7)

$A \to aA \mid a$ (9)

$C \to cC \mid c$ (10)

6	7	8	9	10
bc	c	ab	a	c

2. $S \to AB$

$A \to aAb \mid \lambda$

$B \to bBa \mid \lambda$

a) $S \to AB \mid A \mid B$

$A \to aAb \mid ab$

$B \to bBa \mid ba$

b) $S \to AB \mid aAb \mid ab \mid bBa \mid ba$ (1, 2, 3, 4, 5)

$A \to aAb \mid ab$ (6)

$B \to bBa \mid ba$ (7)

1	2	3	6
aAb	ab	bBa	ab

(3) $S \to YbY \mid ZaZ \mid Yb \mid bY \mid b$

$Z \to aZbZ \mid bZaZ \mid aZ \mid ab$

$Y \to aYbY \mid YaY \mid bY$

$X \to aXb \mid aX \mid Xb$

Símbolos no generadores

Símbolos inútiles (no alcanzables)

Y es alcanzable no generadora (no termina a símbolos terminales)

(4) $L \to \underset{4\ 5\ 6}{aSc} \mid \underset{7\ 8}{Y} \mid \overset{1\ 2\ 3}{AX}$

$Y \to bYc \mid Yc \mid c$

$X \to bXc \mid \underset{10\ 11}{\lambda}$

$A \to \underset{9}{aA} \mid Ab \mid ab$

$L = \{a^i b^j c^k \mid k \neq i+j\}$

7	8	9	10	11
Yc	A	A	b	

(5) $L = \{a^i (b+c)^k a^j \mid k \neq i+j \ ; \ i,j,k \geq 0\}$

$S \to \overbrace{AXB}^{1} \mid a\overbrace{WABW}^{2} \mid \overbrace{WABW}^{3}a \mid$

$A \to aAb \mid \overbrace{aAc}^{4} \mid \lambda$

$B \to bBa \mid \overbrace{cBa}^{5} \mid \lambda$

$X \to bX \mid \overbrace{Xc}^{6} \mid b \mid c$

$W \to aW \mid \lambda$

2	3	5
WABW	WABW	cBa

6 $L = \{ a^i b^j c^k a^m b^p \mid i = j + k \text{ y } m = p \}$

$S \to \overset{1}{XY}$

$X \to \overset{2}{aXc} \mid Y$

$Y \to \overset{3}{aYb} \mid \lambda$

1	2	3
XY	aXc	aYb

0 if $i + j + k = m$

7 $L = \{ a^i b^j c^k d^m \mid i + k = j + m, \; i, j, k \geq 0 \}$

$S \to aSd \mid RRT \mid X$

$P \to aPb \mid \lambda$

$R \to bRC \mid \lambda$

$T \to cTd \mid \lambda$

$X \to bXd \mid T$

1	2	3	4	5
aSd	RRT	X	λ	T